

SPEC-001 v1.4.1 Technical Briefing

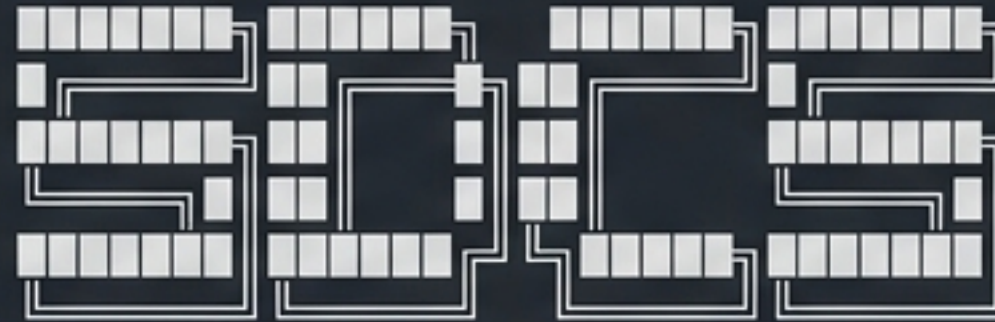
Deterministic Cybernetic Architecture for Autonomous Coding Agents

[SYSTEM BOOT]

Initializing SDCS v1.4.1 ... [OK]

Loading Cybernetic Ontology ... [OK]

Hydrating Teleological Anchor ... [OK]



>> SPEC-DRIVEN COGNITIVE SCAFFOLDING // ONLINE

Transition from probabilistic loops to deterministic control systems.

Externalized working memory and cryptographic test verification.

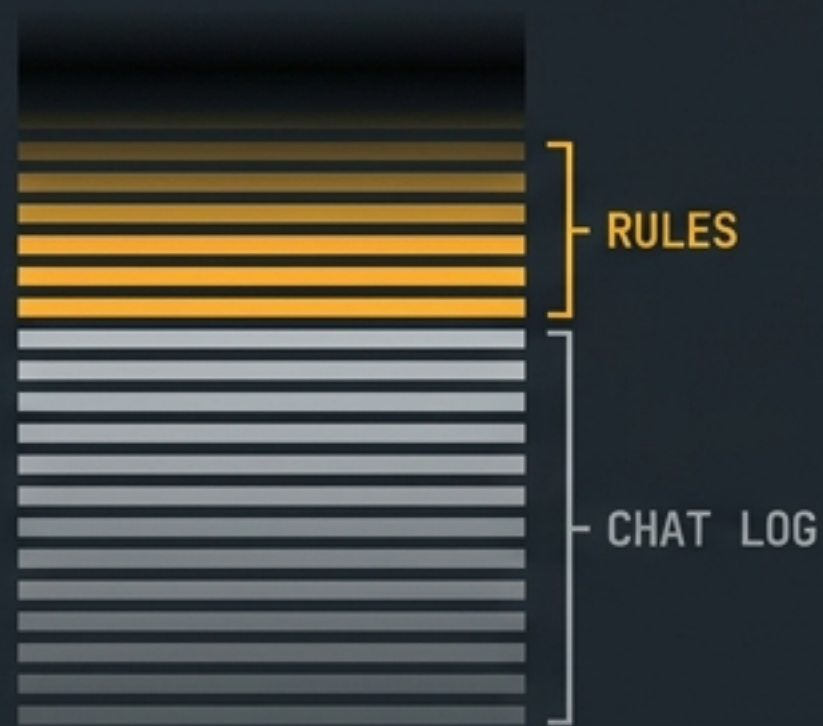
Absolute enforcement of architectural boundaries via AST static analysis.

THE STRUCTURAL PATHOLOGY OF MONOLITHIC PROMPTING

[FATAL: CONTEXT CONFLATION DETECTED] // CANNOT DECOUPLE AXIOMS FROM EPHEMERAL SCRATCHPAD

01 CONTEXT DRIFT

Invariant rules pushed out of effective attention by chat log bloat. Teleology lost.



02 HALLUCINATION

Agent lacks a structural map. Executes recursive brute-force search & burns tokens.



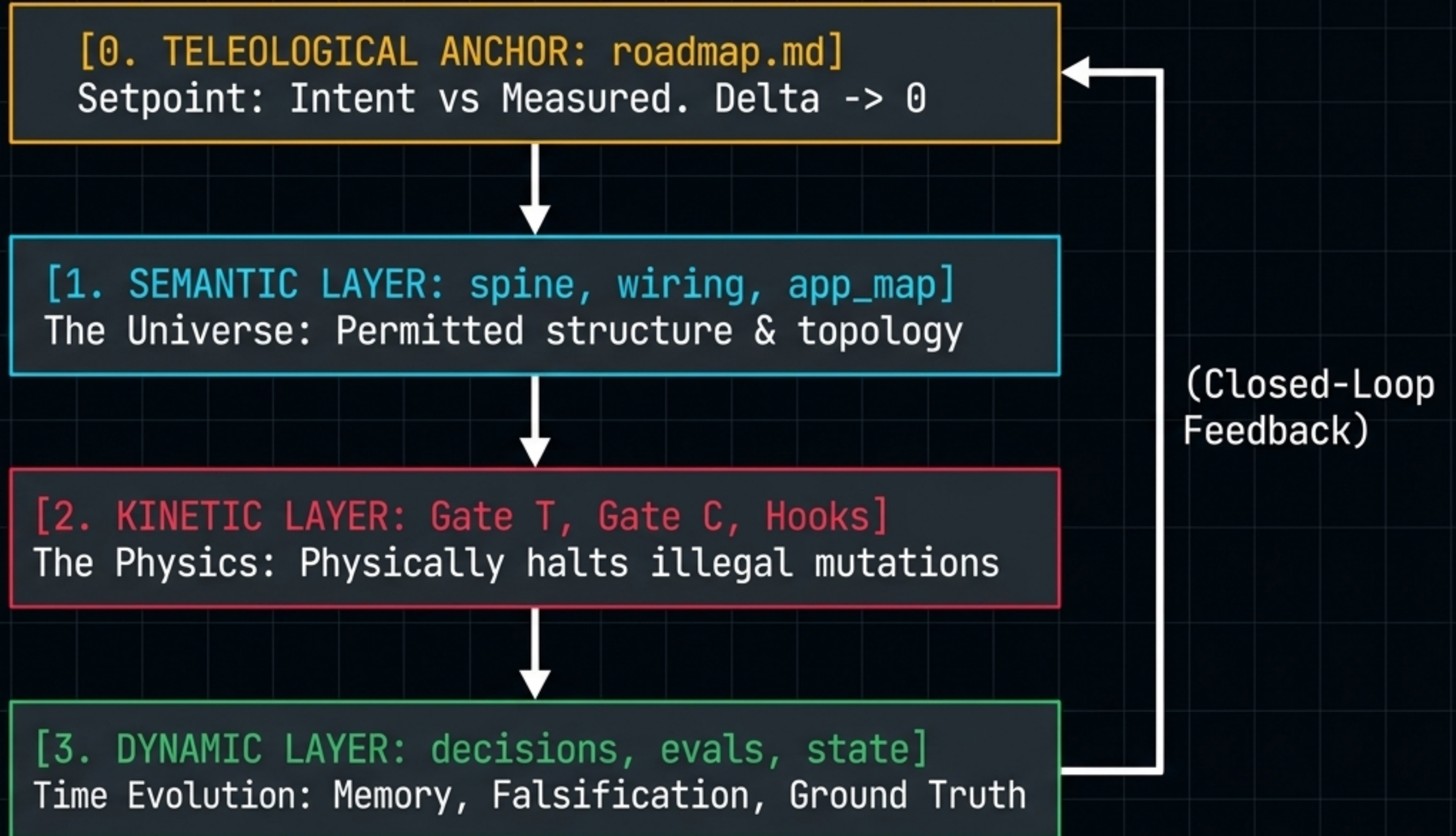
03 EPISODIC AMNESIA

Agent has no negative memory. Repeatedly retries rejected code across rollbacks.



MODELING THE CODEBASE AS A CYBERNETIC CONTROL LOOP

Teleological Anchor + Semantic, Kinetic, Dynamic Layers



EXTERNALIZING COGNITION INTO VERSION-CONTROLLED CONSTRAINTS

The 7-Pillar Cognitive Topography

#	ARTIFACT	PILLAR ROLE	DESCRIPTION
1	spine.md	The Law	Immutable domain axioms & gates.
2	wiring.yaml	The Mesh	Declared interfaces & blast radii.
3	roadmap.md	The North Star	[INTENT] vs [MEASURED] contract.
4	state.md	The Blackboard	Active working memory (≤ 300 tkns).
5	app_map.md	The Compass	Repository index/page table.
6	decisions.md	The Graveyard	Append-only measured rejections.
7	evals.md	The Gcorecard	SHA-256 fixture locks & standing.
8	evals.md	The Scorecard	
+	sessions/*.md	Flight Recorder	Immutable post-shift handoff logs. Never ingested on system boot.

The Strict Epistemological Sequence of Turn 1 Boot

1. spine.md -> (What is strictly forbidden?)



2. roadmap.md -> (What is the active delta?)



3. app_map.md -> (Where do the files live?)



4. decisions.md -> (What hypotheses already failed?)



5. evals.md -> (What is the cryptographic baseline?)



6. state.md -> (What was I doing 10 seconds ago?)



7. wiring.yaml -> (What are the dependency boundaries?)

[!] CRITICAL INVARIANT [!]

**NEVER HYDRATE sessions/*.md
ON BOOT.**

Auto-loading historical prose
recreating context drift.
Querying historical session
logs is strictly on-demand for
forensic debugging only.

Physical Barriers Against Argumentative Rationalization

Kinetic Gate Defenses & Execution Layer

[GATE T] Topological AST Boundary Audit

>> import storage from auth...

[REJECTED] -> Serialized to decisions.md

[GATE M] Cartography Drift Check (sdcs map --check)

>> Orphaned file detected. Run sdcs map --sync...

[REJECTED]

[GATE C] Constitutional Immutability

>> Mutation detected in spine.md without override...

[REJECTED]

[GATE A] Working Memory Budget (sdcs verify --state)

>> state.md budget <= 300 tokens...

[PASSED]

Eliminating Monorepo Context Dilution

Hierarchical Subsystem Cartography Slicing (sdcs map -s)

```
root/  
|- services/  
|   |- auth/  
|   |   |- ... (150 files)  
|- storage/  
|   |- ... (300 files)  
|- proxy/  
    |- router.py
```

```
$ sdcs map --subsystem proxy  
[EXTRACTING FOCUSED CARTOGRAPHY SLICE]  
  
root/  
|- services/ [PRUNED]  
|- proxy/ [MAPPED]  
|   |- app_map.md ← Paging sub-table...  
|   |- router.py [ACTIVE]  
|   |- middleware.py [ACTIVE]  
|- storage/ [PRUNED]  
  
>> Token consumption reduced by 85%.
```

Enforcing Symmetric Episodic Memory

Negative Memory vs. Positive Ground Truth

[PILLAR 6: THE GRAVEYARD]

File: decisions.md

State: **NEGATIVE** (Rejections)

REJ-014

CLAIM: 5x5 closing filter.

MEASUREMENT: Precision dropped
from 86% -> 43%.

REOPEN: DPI drops below 150.

>> Agent is structurally blocked from
cyclical regressions.

[PILLAR 7: THE SCORECARD]

File: evals.md

State: **POSITIVE** (Empirical)

ID	Digest (SHA-256)
TC-01	a9f4c21... [OK]
TC-02	b3d8e9f... [OK]
TC-03	a9f4c21... [ERR!]

>> Duplicate hashes physically halt
Phantom Corpus drift.

Atomic Baseline Recalibration and Integrity

One-Pass Cryptographic Hashes (sdcs eval record)

```
$ sdcs eval record TC-001
[RECALIBRATING FIXTURE TC-001]
Target: tests/fixtures/vector_001.json
Target: tests/fixtures/vector_001.json
Normalizing whitespace & comments... [DONE]
Computing SHA-256 digest... [DONE]
Writing to evals.md markdown table... [ATOMIC SUCCESS]
>> TC-001: 7f83b1657ff1fc53b92dc18148a1d65dfc2d4b1fa3d677284add200126d9069

$ sdcs eval record all
>> Recalibrated 42 registered fixtures successfully.
```

Canonical Normalization: Strips trivial whitespace and dummy comments before hashing to defeat evasion tactics.

Corpus Diversity Invariant: Forces genuine benchmark variation over repetitive dummy mocks.

Eradicating Compaction Amnesia

Lossless Compaction Protocol & Offloading Context to Disk

[PREPARE FOR COMPACT // TRIGGERED AT 75% SATURATION]

1. Topology Audit -> `sdcs verify --topology`

2. Flight Recorder -> Write immutable sessions/YYYY-MM-DD.md

3. Blackboard Prune -> Overwrite state.md strictly ≤ 300 tkns

4. Episodic Sweeps -> Sync decisions.md & app_map.md

5. EMIT SIGNAL: Ready for compaction.

[WIRING MUTATION INVARIANT]

New modules must be declared in wiring.yaml in-stride.
Bypassing Gate T boundaries prior to memory wipe is strictly forbidden.

The Tactical Tooling Suite for Production Execution

The SDCS CLI Runtime Harness

CLI COMMAND	DETERMINISTIC SYSTEM IMPACT
<code>sdcs init</code>	Bootstraps 7 pillars + index + hook config
<code>sdcs verify --topology</code>	Gate T AST audit -> appends REJ to graveyard
<code>sdcs verify --state</code>	Gate A linter enforces 350-token hard limit
<code>sdcs map --sync</code>	Reconciles disk state with cartography map
<code>sdcs session index</code>	Parses shift history into manifest.jsonl
<code>sdcs graph --ascii</code>	Renders wiring.yaml DAG to terminal stdout
<code>python sdcs_init.py --grill</code>	Adversarial spec elicitation hardens requirements

- Zero-dependency Python package installable in any repository.

Achieving Zero-Drift Engineering via SDCS

Production Readiness & System Deployment

[SYSTEM DEPLOYMENT AUTHORIZED]

```
$ pip install sdcs  
$ cd your-repository  
$ sdcs init  
$ git config sdcs.mode strict
```

STATUS: ACTIVE. COGNITIVE SCAFFOLDING ENGAGED.

Agent cognition decoupled. Invariants locked on disk.
Cybernetic control loop closed.

>> Awaiting Phase 1 Orientation... █